

Causes of variation in vaccine responsiveness in a wild rodent population

Wild vaccinology: determining the causes of variation in vaccines in a lab-to-wild rodent population

Causes of vaccine failure: bridging the translational divide using a lab-to-wild model

Causes of low vaccine responsiveness: bridging the translational divide using a lab-to-wild model

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ABSTRACT

Most vaccines fail to reach their intended target population, typically by failing late into the clinical trial pipeline. Even established vaccines may lose efficacy in certain individuals or groups that differ substantially from those in which they were initially tested, leading to breakthrough infections and the inability to achieve or maintain herd immunity. However, the causes of vaccine hyporesponsiveness remain difficult to identify, quantify, and therefore pre-empt. We hypothesise that the leading causes of vaccine hyporesponsiveness are environmental, and are exacerbated when individuals are required to allocate limited resources to growth, reproduction and immunity. Here, we show that circulating vaccine-specific antibodies dropped by 50% in a wild population relative to individuals of the same species that had been reared in laboratory conditions. Using paired cohorts of laboratory-reared and wild wood mice (*Apodemus sylvaticus*) vaccinated with a single and double-dose diphtheria toxoid vaccine (with adjuvant alum), we found that this loss in vaccine responsiveness was explained in the most part by diet, with weaker negative impacts of being male and reproductively active. However, helminth infection did not directly suppress vaccine-specific antibody production once adjusted for habitat and time since immunisation. While much variation remains to be explained, our results indicate that the environment plays a substantial role in vaccine failure, notably but not exclusively via nutrition, and that laboratory settings may systematically overestimate vaccine performance. We anticipate that our study inaugurates a robust method for disentangling and quantifying the complex factors that cause vaccine responsiveness in the wild. It may also call for a need to systematically raise efficacy thresholds for initiating clinical phases of vaccine development, and suggests that simple dietary interventions might improve individual responsiveness to immunisation.

INTRODUCTION

Unanticipated variability in individuals' responses to vaccination can lead to a lack of protection from the target pathogen, continued disease transmission, and an increased risk of vaccine escape¹⁻⁴. This heterogeneity is largely driven by an individual's pre-existing immune state, as demonstrated by the association between innate inflammatory markers in patients' peripheral blood before vaccination and subsequent antibody responses to different vaccines⁵. Vaccine efficacy can further be affected by genetics⁶, sex⁷, reproductive status⁸, past and current infections⁹⁻¹¹, and nutrition¹²⁻¹⁴. These factors can combine to affect both adaptive and innate immunity, and therefore, disease and transmission¹⁵. However, it is unclear how these factors interact and how to mitigate their effects to maximise vaccination efficacy. The limitations to our understanding of variability in vaccine efficacy stem in part from the inadequacy of current pre-clinical models used in vaccine development, which typically utilise defined inbred, non-reproductive, single sex, age-controlled animals that are bred in captivity under strictly controlled pathogen-free conditions. As a result, only a minority of vaccines, ~6%, successfully overcome their transition to "the wild" and reach mass market¹⁶. Moreover, even successful vaccines can suffer from substantial variability between individuals and groups, notably for people of different ancestry, e.g. in response to measles vaccination¹⁷⁻¹⁹, or in urban *vs.* rural populations where differences have been reported in the expression of BCG- and tetanus-specific cytokine recall and antibody responses^{20,21}.

To increase the generalisability of rodent models of immunology and vaccinology, recent efforts have been made to integrate natural variability into immunological research²²⁻²⁶, but there is still uncertainty about how this heterogeneity affects the innate and adaptive arms of the immune system, and what the consequences might be for vaccine responsiveness. More specifically, it is unclear how environmental factors and past stressors influence organisms' responses to immunisation. This complexity blurs the causal relationships

between environmental and immunological processes, making the design and interpretation of *in vivo* vaccination experiments difficult and their relevance to their intended target populations intractable.

Here we propose a structural causal model (SCM)²⁷ to predict the effects of environmental variation on vaccine responsiveness. This causal framework, which utilises graphical models to inform adjustment strategies^{28,29}, can eliminate or reduce multiple sources of confounding, such as common causes (misattribution of causal effects), sampling biases (non-representative test population), and transportability biases (lack of generalisability). We performed an immunisation study using a coupled laboratory-wild wood mouse (*Apodemus sylvaticus*) system, which allows longitudinal sampling as well as experimental manipulation in both controlled laboratory conditions and in their natural woodland habitat. This allowed us to use the same assays and interventions across habitats and populations, and to replicate and condense the lab-to-clinic vaccine test-translate pipeline. We then compared the generation of vaccine-specific antibody responses to an alum-adjuvanted diphtheria toxoid vaccine (DTV) between laboratory and natural settings and between two diets (supplemented and control) after vaccination. We have previously demonstrated that this high quality, whole diet supplementation drives a sustained reduction in helminth (*Heligmosomoides polygyrus*) burdens and transmission, while also improving the efficacy of anthelmintic treatment³⁰. However, reduction in helminth burdens can lead to increased coccidial infection³¹. How these dynamics are mediated and might interact with responses to vaccination is therefore unknown. We predicted that vaccine responses in wild mice should be lower than in lab wood mice, higher in diet-supplemented animals, and thus that diet might “rescue” hyporesponsiveness in wild rodents.

Then, we used Bayesian SCMs to quantify the physiological and environmental direct and indirect causal effects of variation in the ability of recipients to generate vaccine-specific

[SAB] Is the prediction useful here? too contrived?

[SV] We will have to add more background regarding diet and vaccines to justify our predictions >>>Enough?

antibody responses (vaccine responsiveness), to identify the main causes of loss of vaccine efficacy when translating a vaccine from the laboratory to the wild, and to accurately model how uncertainty and biological variability propagate along the causal network (see casual flow diagram, Fig. S1).

RESULTS

Vaccination induced stronger but variable antibody responses in the laboratory than in the wild

To assess how a laboratory environment, enriched diet, and vaccination protocol affect vaccine responsiveness, we designed a paired cross-factorial study in wild and laboratory *A. sylvaticus* in which mice were provided access to high or low-quality diet, and then immunised with a diphtheria-alum vaccine (DTV) in a single or double shot (see details in Materials, Methods, and Models; Figure 1).

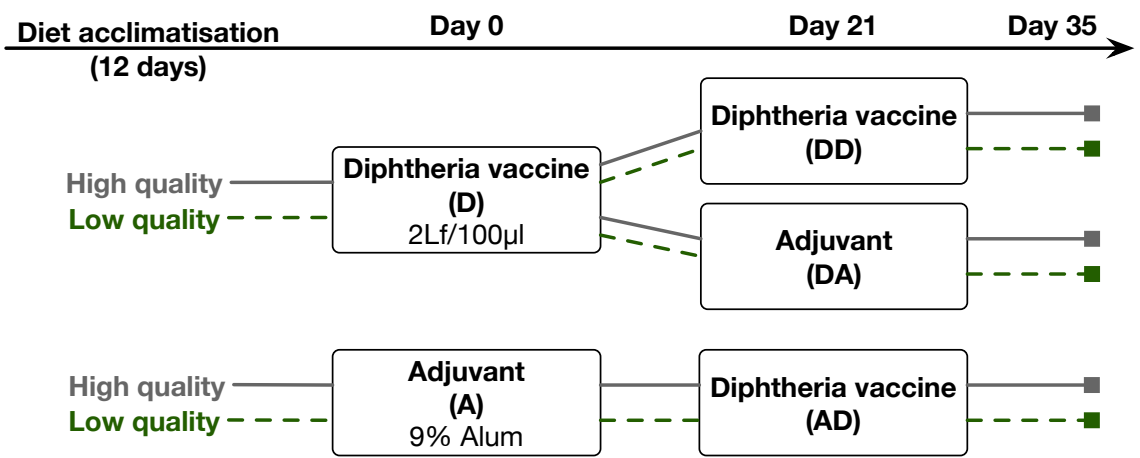


Fig. 1: Allocation of diet and vaccines. Wood mice from a laboratory colony, and from the wild were provided enriched (TransBreed) or standard (lab: normal chow; wild: no supplementation) diets. After at least 12 days, mice were allocated at random to immunisation with a 2Lf/100µl Diphtheria Toxoid + 9% alum vaccine (D) or with adjuvant alone (A). Twenty-one days after their first immunisation, mice that had been immunised with D were then given a second shot of the vaccine (DD) or adjuvant alone (DA), while mice that had received an initial adjuvant vaccine (A) were all given D for their second shot (AD).

DTV-specific IgG1 varied substantially within and between wild and laboratory-bred wood mice, immunisation protocols, and diets. Group-wise comparisons revealed substantially lower vaccine responsiveness in wild *vs.* laboratory animals and in dietary

[SV] Perhaps a schematic here summarising the system and the approaches (experiment + SCM) used in this study would be useful here?
>>>See Fig. S1; does this help?

supplemented *vs.* individuals, as well as substantial variability across all immunisation regimens (Figure 2A). Only mice that were administered the DT antigen generated DTV-specific IgG1, indicating that there was no pre-existing immunity to this antigen in our study populations (Figure 2A, 'A'). Immunisation was most effective with two doses, partially rescuing responsiveness in the wild mice (LRT of Habitat $\times$ vaccine regimen: $\Delta dof = 4, \chi^2 = 30.6, p < 0.001$; Figure 2A, 'DD'), while only lab mice mounted similar antibody levels with a single dose if given enough time (Figure 2A, 'DA').

To further quantify how habitat (lab *vs.* wild), diet (supplemented *vs.* control), and immunisation protocol correlated with variation in vaccine responsiveness, we constructed a multilevel Bayesian model of those factors with intercepts varying by individual and immunisation protocol. Mice in the wild generated substantially weaker antibody responses to DTV than lab-based animals (Figure 2B, 'Hab'). Surprisingly, diet supplementation reduced antibody levels across all immunisation protocols relative to individuals on standard diet – standard chow for lab mice; natural foraging for wild mice – (Figure 2B, 'Diet').

In summary, the model predicts an average loss of DTV responsiveness of $65\% \pm 8\%$ when moving from the lab to the wild, and of $26\% \pm 7\%$ on the TransBreed diet, while a booster immunisation ('DD') roughly doubled the production of DT-specific antibodies relative to a single dose during the observation period of this experiment (Table 1).

[SAB] Calculate contrasts for E under H and D

Construction of a structural causal model of vaccine responsiveness

Next, to understand more specifically the relative contributions of diet and habitat to vaccine responsiveness, and how that effect might be mediated by parasite infection, reproductive status, and body condition, we built a causally-explicit model of the hypothesised causal relationships between (Figure 3).

To do this, we constructed a directed acyclic graph (DAG, Figure 3) to represent the hypothesised structure of the causal model $\mathcal{C}$ linking our variables of interest to vaccine

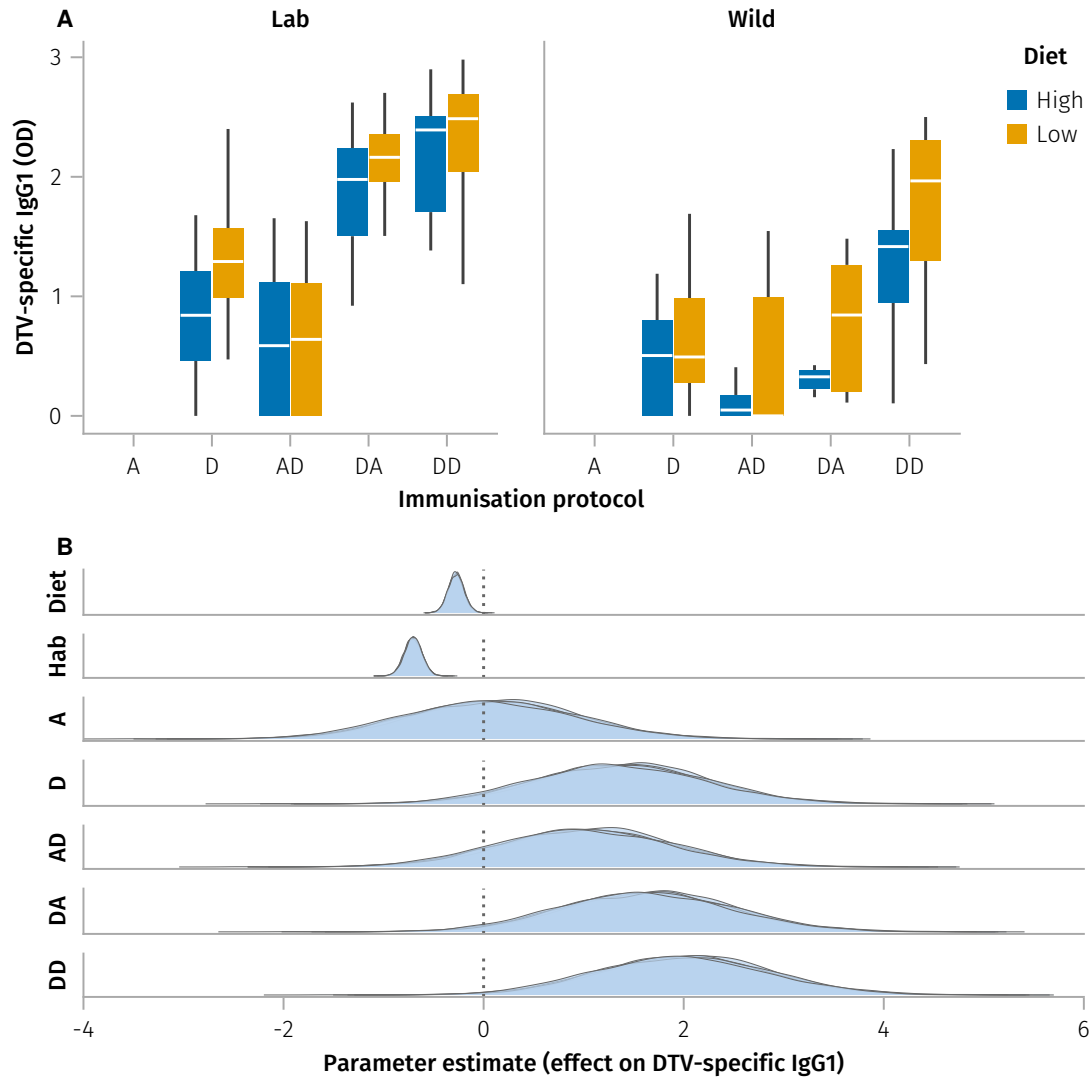


Fig. 2: Vaccine-induced DTV-specific IgG1. **A**, diphtheria toxoid vaccine (DTV)-specific IgG1 optical density in mice from laboratory- and wild-habitats offered high and low quality dietary supplements. Mice were immunised either once, with adjuvant as control (A) or DTV (D) only, or twice (i.e. with booster), using adjuvant alone followed by DTV at subsequent recapture ≥ 7 days after the previous immunisation (AD), DTV and subsequently adjuvant alone (DA), or two separate doses of DTV (DD). Only mice immunised at least 7 days prior to antibody measurement are shown, and each point represents one of 1–4 recaptures for each individual. **B**, distribution of posterior distributions of the coefficients for diet, habitat, and vaccine formulation (A, D, AD, DA, DD) of a multilevel Bayesian model adjusted for repeated individual measures. Each density curve represents the distribution of 3,000 samples from one of four MCMC chains (see Fig S2).

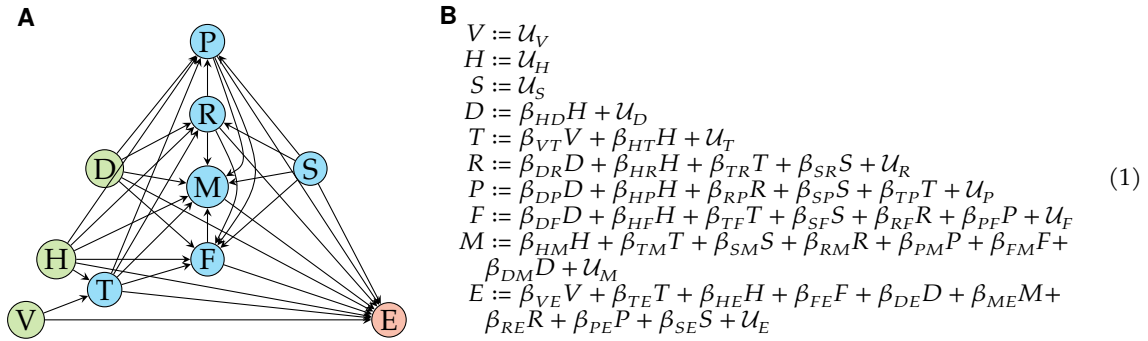


Fig. 3: Graphical Causal Model of vaccine efficacy. **A**, Directed acyclic graph representing hypothesised causal effects expressed by the SCM $\mathcal{C}$ (Model 1, right) driving variation in efficacy E of vaccine V in laboratory or wild mice H , under supplemented or control diet D (green vertices depict those that were experimentally intervened upon). Covariates and potential confounders (blue vertices) were time since vaccination T , parasite infection (presence or burden) P , reproductive status R , body mass M , fat scores F , and sex S . Directed edges depict causal effects hypothesised to operate between vertices. **B**, Assignments derived from the DAG with coefficients β_{ij} representing the effect of cause i on effect j . $\mathcal{U}_j$ represents the unmeasured (residual) variation of the response variable.

efficacy E and oriented the causal effects based on the following criteria. Vaccination V being allocated at random, it has no parent node. Its effects were hypothesised to always be mediated by time since vaccination T to account for the dynamics of antibody production accumulation within each individual. A mouse being born in the lab or in the wild habitat H was considered random, and therefore took no causal input but was hypothesised to potentially affect all other variables except V and sex S . Diet D was allocated at random in the laboratory colony, but because wild mice could also forage freely, we considered that being born in the wild could have a causal effect on diet $H \rightarrow D$, specifically on the effect of the supplementation relative to the control laboratory or wild diet. Diet was hypothesised to affect reproductive status R , body mass M , body fat F and parasite burdens P as well as E , with fat scores F hypothesised to be a cause of variation in body mass M . Sex S being genetically determined and therefore subject to Mendelian randomisation³², and all treatments being randomised with respect to sex, S was considered causally prior to reproductive status R , body mass M , body fat F and parasite burdens P . Reproductive status R being driven by sex, it was considered causally prior to body mass M and parasite

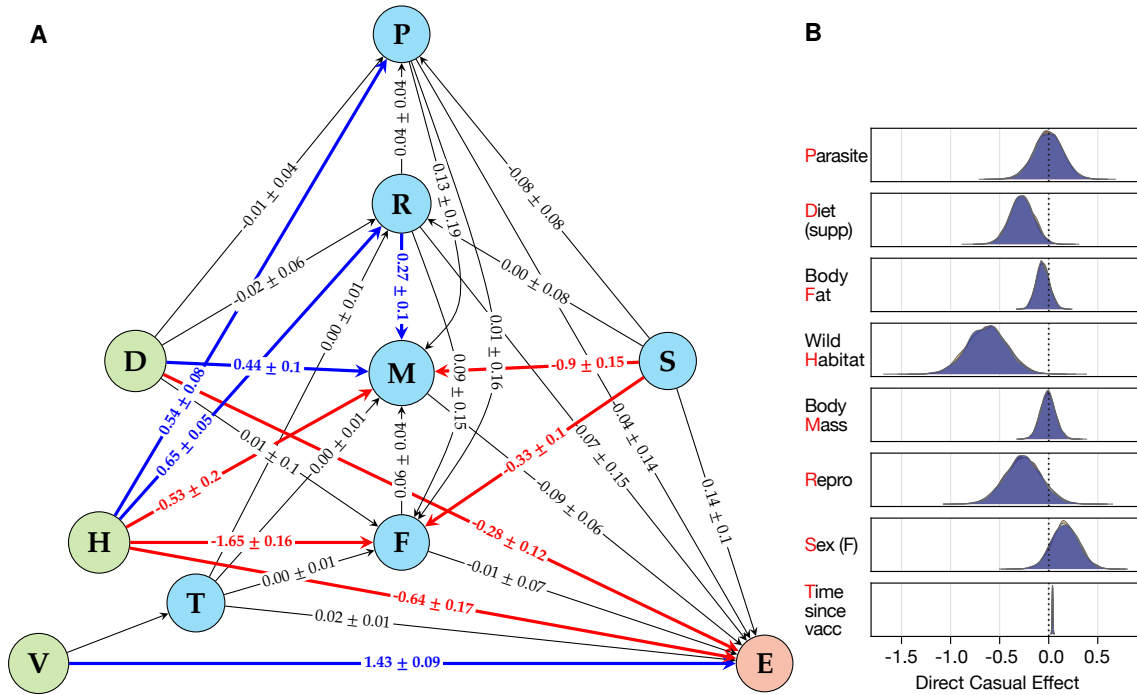


Fig. 4: Fully resolved Causal Model of vaccine efficacy. **A**, Directed acyclic graph representing point estimates of positive (blue) and negative (red) direct causal effects $\pm$ std driving vaccine responsiveness E . **B**, MCMC sampling of posterior distributions of the coefficients for $E := \beta_{DE}D + \beta_{FE}F + \beta_{HE}H + \beta_{ME}M + \beta_{PE}P + \beta_{RE}R + \beta_{SE}S + \beta_{TE}T$ (see Model parameterisation methods).

burdens P . Vaccine efficacy E was hypothesised to be potentially affected by all other variables.

We then tested the validity of the causal hypotheses described above using the DAG to construct each model. For example, the DAG in Fig. 3 implies that parasite infection should be independent of vaccination if we adjust for diet, reproductive status, sex, and habitat ($P \perp\!\!\!\perp V \mid D, R, S, H$); we therefore fit a binomial generalised linear model $glm(P \sim \alpha + \beta_V V + \beta_D D + \beta_R R + \beta_S S + \beta_H H)$. This process was repeated for all implied consistencies derived from the causal graph (see Model validation), and were found consistent with the observed data.

Finally, we fully parametrised $\mathfrak{C}$ (Model 1) using a set of Bayesian linear models containing variables required to ensure unconfounded estimation of the total and natural direct causal effects operating between the nodes of $\mathfrak{C}$ (Fig. 4).

[SAB] Complete values of edges

Main drivers of vaccine efficacy

[SAB] Contrasts!

Total and direct causal effects of the wild on vaccine responsiveness

DISCUSSION

Weak / variable effect of worms^{10,11,33}.

Effect of nutrition¹⁴

MATERIALS, METHODS, AND MODELS

All animal work was conducted in accordance to UK Home Office in compliance with the Animals (Scientific Procedures) Act, 1986. Both laboratory and field experiments were approved by the University of Edinburgh Ethical Review Committee and carried out under the UK Home Office Project Licence 70/8543. The dosage of diphtheria vaccine was trialled in laboratory reared wood-mice before the experiments for safety and efficacy. Animal sacrifice was performed using appropriate Schedule 1 Methods. Fieldwork was carried out with permission of the Forestry Commission Scotland under the permit SUR09.

See Supp methods

[ABP] What do we plan to include in the supplementary methods?

Laboratory wood mouse experiment

The laboratory experiment was conducted on a wild-derived but now captive, outbred wood mice colony that has been bred and maintained at the University of Edinburgh³⁰ (See SI for more details). The experiment was carried out in two replicate blocks, with 36 animals in each block (18 males, 18 females). At 12 days before the start of the experiment (d-12), all mice were shifted to new diet regimes and given time to acclimatise; half of the animals in each block (9 males and 9 females) were given TransBreedTM, a high-quality whole-diet mouse chow. We have previously shown that wood mice supplemented with TransBreedTM were more resistant to the gastrointestinal nematode *Heligmosomoides polygyrus*, cleared worms more effectively after anthelmintic treatment and produced

stronger general (total IgA) and parasite-specific (IgG) immune responses in both wild and laboratory conditions³⁰. The other half of the mice in this experiment were fed RM1 (Rat and Mouse Maintenance), a lower quality diet, but commonly used maintenance mouse chow which supports maintenance. Both TransBreedTM and RM1 are standard laboratory diets, with TransBreedTM containing higher amounts of starch, protein, fat and micro-nutrients compared to RM1 (see SI for detailed nutritional compositions). Within each diet regime, both male and female animals were randomly assigned to three experimental groups (Fig. 1). Animals belonging to Groups 1 & 2 (n = 24) were subcutaneously injected with 100µl of diphtheria vaccine (2 Lf per 100µl of adjuvant (9% Potassium Aluminium Sulphate) on d0. The vaccine was prepared in our laboratory one day before administration, using vaccine-grade diphtheria inactivated toxoid protein (Alpha Diagnostic International) adsorbed to the adjuvant (9% Alum). Animals assigned to Group 3 (n = 12) were injected with a control, referred to as 'adjuvant'; 100µl of adjuvant (9% Potassium Aluminium Sulphate) suspended in 1X PBS on d0. Small volume blood samples were taken via tail snip 14 and 17 days after the first injections (d14 and d17) to measure the primary antibody response. Twenty-one days (d21) after the first injections, animals in Group 1 (n = 12) were injected again with 100µl of the diphtheria vaccine (referred to as the 'booster'), whereas those assigned to Group 2 (n = 12) were injected with the control dose as described above (referred to as 'adjuvant'). Animals assigned to Group 3 (n = 12) that had previously been injected with the control dose, now received an injection of 100µl diphtheria vaccine. Another blood sample (20- 50µl) was taken a day later (d22) via cheek venipuncture. All animals were sacrificed 14 days after the second injection (d35) and a blood sample for measuring the secondary antibody response was taken. The same procedures and timeline were carried out for the second experimental block of animals (n = 36). Mice were co-housed throughout the experiment, in same sex groups of three, with equal representation of the three experimental groups in each cage.

Wild wood mice experiment

We conducted a 10-week field experiment in a natural population of wood mice with a design similar to that of the laboratory experiment. The experiment took place in a woodland in Falkirk, Scotland, UK (Callendar Wood, 55.990470, -3.766636), where we established four trapping grids (60 m X 40 m per grid) with 10m spacing between each of the 35 trapping stations. At each station, we set a pair of Sherman live traps (H.B. Sherman 2X2.5X6.5-inch folding trap, Tallahassee, FL, USA). We randomly selected two of the four grids to be 'supplemented'; 12 days before we began live-trapping, we started supplementing these grids with extra resources. Specifically, we scattered 6 kg TransBreed™ (high quality, whole- diet mouse food) pellets evenly around the grid, and this supplementation continued throughout the experiment, twice per week. The two 'control' grids did not receive any supplementation. Wood mice on all four grids had access to their normal diet and food items, so this extra resources on the 'supplemented' grids served as an additional high nutritional food resource.

From July-September 2018, wood mice were trapped 2–3 nights per week using live traps baited with grains, carrots, and bedding. Additionally, traps on the supplemented grids were also baited with 1-2 TransBreed™ pellets. For identification, all newly captured mice weighing >13g were subcutaneously tagged with a unique 9-digit microchip passive induced transponder (PIT tag; FriendChip AVID2028, Norco, CA, USA). On first capture, mice were assigned to one of the three vaccine treatment groups described above in the laboratory experiment; mice assigned to these groups received the same primary and booster doses of the diphtheria vaccine or control injections as described below (see Figure 1). Throughout the experiment, weekly small volume blood samples were taken via tail snip from each tagged animal to measure their antibody response to the diphtheria immunisation. Animals recaptured 14 or more days after their second injection were sacrificed and a terminal bleed was collected to measure their antibody response and

[SV] Could refer to ARS's work that shows reduced helminth burdens in diet-supplemented wild mice

[ABP] Agreed. I added a few more details above - but let me know if you think we need to add more

adult *Heligmosomoides polygyrus* worm burdens. *H. polygyrus* is a natural gastrointestinal nematode found at high prevalence (20-100 %) in wild populations of *A. sylvaticus*^{30,34–36}, as well as being a well-studied model system of human gastrointestinal nematodes where it has been found to be highly immunomodulatory^{37,38}.

Additionally, at all captures, we took the following demographic metrics for each individual: sex, body condition, reproductive condition, weight (g), and length (mm). Sex and reproductive condition of each mouse was assigned by examining the genitals, with males having larger urogenital gap compared to females. Males were classified as reproductively active if their testes had visibly descended to scrotal sacs, and females were classified as reproductively active if they had a perforated vagina or were visibly pregnant or lactating. Body condition was measured by assigning dorsal and pelvic fat scores on a qualitative scale from 1-5, where 1 represented a very low condition mouse, while a 5 represented a mouse with ample fat reserves. This was accrued out by palpating the back and pubic bones³⁹.

Vaccine responsiveness

Vaccine responsiveness was measured as the concentration of anti-diphtheria IgG1 antibodies; this is the gold standard method used in vaccinology to measure the antibody-specific response to immunisation and is used specifically for the diphtheria toxoid vaccine used here⁴⁰. To measure diphtheria vaccine responsiveness, we collected blood samples from both the laboratory and the wild mice and these were centrifuged (@24,000 rpm for 10 min) within 4–6 hours of collection. The sera were separated from the pellets and stored at -80 °C until further analysis. We used ELISAs adapted for diphtheria-specific antibodies from Clerc et al. (2019a), to quantify Anti-diphtheria IgG1 antibodies in the sera samples. 96-well plates (Nunc™ MicroWell™) were coated overnight at 4 °C with diphtheria toxin (2µg/mL) diluted in carbonate buffer (50 µl per well). After washing the plates with TBS (10 X) and Tween80 thrice, 100 µl of TBS(1X)-4percent BSA was added per well

and incubated at 37 °C for 2 hours to block non-specific binding sites. Plates were then tapped dry and 50µl of serum samples serially diluted (starting dilution 1:100) in TBS (1X)-4percent BSA buffer were added per well and left at 4 °C for binding overnight. The next day, 50µl HRP-conjugated anti-mouse IgG1 detection antibody (Southern BioTech) was added per well after washing the plates 4 times with TBS (10X) and Tween80 and tapping them dry. After incubation at 37 °C for 1 hour, the plates were washed again, first 4 times with TBS (10X) and Tween80 and then twice with dH2O. 50µl of TMB substrate solution was added per well and the enzymatic reaction was left to develop in the dark for 7 min. The reaction was stopped after 7 min using 50µl sulphuric acid (0.18 M) per well. Absorbance at 450 nm was recorded using an ELISA plate reader (Multiscan, Ascent Labsystems) immediately after. For each sample, blank-centred optical densities (OD) of three consecutive wells (of dilutions 1:3200, 1:6400, 1:12800) were averaged to obtain diphtheria-specific IgG antibody concentration.

Data processing and analyses

DT-specific antibody ELISA optical densities (E) were $1 + \log_{10}$ -transformed and, body weight (M), fat scores (F) and days post-vaccination ($T \geq 1$), were Z score-transformed to help the selection of sensible priors and to improve computational stability. All models were adjusted for time T between vaccination and blood collection to account for natural kinetics of the antibody response. Fat scores ranged from 2 to 5 and were derived from the sum of two five-point scales for pelvic and dorsal fat, where 1 is ‘bony’ to palpation and 5 is ‘fat’, and were treated as continuous for the analysis. All other variables were treated as binary.

All data processing was performed in Julia v1.9⁴¹ using packages CSV.jl v0.10⁴² and DataFrames.jl v1.3⁴³.

Generalised linear mixed models of the effects of experimental interventions on habitat, diet, and vaccine formulation

Generalised linear mixed models (GLMM) were used to estimate the effects of habitat, diet and vaccine formulation on vaccine-specific antibody concentrations measured as optical densities read-outs. Mouse ID was modelled as a random effect to account for multiple testing. The interaction between vaccine regimen and habitat was included in the full model to test whether mice in the wild would respond differently to vaccine timing and boosting than laboratory mice. Likelihood ratio tests (LRT) were used to test the contribution of the interaction terms to the model. We used Julia v1.9⁴¹ package MixedModels.jl v4.13⁴⁴ was used for GLMMs and LRTs.

Bayesian statistics: choice of priors and sampling of posterior estimates

We used Bayesian multilevel models to minimise the effects of imbalance on our analyses and to quantify the uncertainty of those estimates given the available data. Priors for intercepts and slopes were chosen to conform to the observed data, with equal means and $2.5 \times$ the observed variance (see Supplementary Data: Multilevel Models). Typically, intercepts were given Gaussian priors of mean $\mu = \text{mean}(y)$ and $2.5 \times \sigma$ those of the observed response variable y while population-level coefficients were given multivariate normal (Gaussian) priors of mean $\mu = 0$ and variance $\sigma = 2$. Population-level residuals were assumed to follow an exponential distribution with scale $\theta = 1/\text{std}(y)$. Hyperparameters for group level intercepts and coefficients were given Gaussian priors and their variances were given positive-truncated Cauchy priors with location $x_0 = 0$ and scale $\gamma = 2$. Posterior estimates were sampled using the Hamiltonian Monte Carlo algorithm No U-Turn Sampler⁴⁵, typically across 4 threads simultaneously with $> 3,000$ draws. Turing.jl⁴⁶ was used for Bayesian modelling; Makie.jl v0.17⁴⁷ was used for plotting.

Structural Causal Models

The structural causal model (SCM) framework^{27,48} proposes explicit qualitative solutions to biases inherent in datasets and can guide the statistical estimation of the mechanisms by which data are generated in complex systems^{49,50}. Further, Bayesian statistical methods for estimating the causal effects indicated by the SCM allows flexibility of imbalanced small data and missing values⁵¹, and provides explicit estimation of uncertainty given assumptions made by the model⁵².

Model construction

Each of the variables X described above was included in the structural causal model $\mathfrak{C}$ with the following assignments of the form $X_j := f(X_{k \neq j}) + \mathcal{U}_j$ where the unmeasured noise variable $\mathcal{U}_j \sim \text{Normal}(0, 4)$, was assumed to be independent and identically distributed, and $f(\cdot)$ was assumed to be linear (Supplementary Data: SCM construction):

A directed acyclic graph (DAG) expressing $\mathfrak{C}$ (Model 1) was used to build unconfounded models for the estimation of causal effects between each variable (Figure 3).

Model construction was performed in Julia v1.9⁴¹ using package Dagitty.jl and Turing.jl⁴⁶, and the DAG corresponding to $\mathfrak{C}$ was drawn in Tikz⁵³.

Model validation

To validate $\mathfrak{C}$ (Model 1), we tested the independencies it implies following chain, fork, and collider rules^{27,54}. Specifically, we used the observed values for each of the nodes of the causal graph (Figure 3) to test the truth of the following 13 implied independences: $(D \perp\!\!\!\perp S)$, $(D \perp\!\!\!\perp T \mid H)$, $(D \perp\!\!\!\perp V)$, $(F \perp\!\!\!\perp V \mid T, H)$, $(H \perp\!\!\!\perp S)$, $(H \perp\!\!\!\perp V)$, $(M \perp\!\!\!\perp V \mid T, H)$, $(P \perp\!\!\!\perp T \mid D, R, S, H)$, $(P \perp\!\!\!\perp V \mid T, H)$, $(P \perp\!\!\!\perp V \mid D, R, S, H)$, $(R \perp\!\!\!\perp V \mid T, H)$, $(S \perp\!\!\!\perp T)$, and $(S \perp\!\!\!\perp V)$. For each of these tests, we constructed a generalised linear mixed model with the appropriate link function, with mouse ID and immunisation protocol as random effects, and considered the statement true if $P_X > 0.1$, where $X \subseteq \mathfrak{C}$ (Supplementary Data: SCM

construction). We used MixedModels.jl v4.13⁴⁴ to test the conditional independences implied by $\mathcal{C}$.

Model parameterisation

Next, we encoded $\mathcal{C}$ as a Bayesian network in Turing.jl⁴⁶ to represent each variable as a probability distribution to be updated by the observed values. This allowed us to capture conditional expected values for each variable and their uncertainty. Posterior estimates were validated with MixedModels.jl⁴⁴.

Missing data. Twenty-two mice had no fat scores, resulting in 67 missing values in the full repeated sampling dataset. To avoid having to remove those mice from the analysis, we used Bayesian imputation to estimate plausible probability distributions of each of the missing values in light of its parents (PA) in $\mathcal{C}$ (Model 1), where $F := f(PA_F) = f(T, H, D, P, S)$ (Figure 3).

$\tilde{F}$ denotes the variable F that contains imputed values; E_i denotes the expected distribution of E given $\tilde{F}$.

Model updating and sampling. All Bayesian modelling was performed in Julia v1.9⁴¹ using package Turing.jl v0.25⁴⁶.

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COMPETING INTERESTS

The authors declare that there are no competing interests.

AUTHOR CONTRIBUTIONS

Conceived and designed the experiments: ABP, SAB

Performed the experiments: SV, AS, ABP, SAB

Analysed the data: SV, SAB

Add details of the ISSF vaccination grant, Saudamini's Darwin Trust funding, ABP's Chancellors Fellowship, Amy Sweeny's Darwin Trust funding, SAB RS and BMGF - what else?

Contributed reagents/materials/analysis tools: ABP, SAB

All the authors reviewed the manuscript.

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TABLES

Table 1: Posterior predictions of model parameters for random intercept Gaussian model $E \sim \text{Normal}(\alpha[ID] + H + D + V, \sigma)$.

Parameter	Mean	Std	5.5%	50%	94.5%
Intercept	-0.082	0.855	-1.465	-0.066	1.266
Diet	-0.265	0.076	-0.387	-0.265	-0.144
Habitat	-0.650	0.085	-0.788	-0.649	-0.515
A	0.026	0.845	-1.297	0.017	1.387
D	1.301	0.845	-0.024	1.294	2.654
AD	0.789	0.846	-0.539	0.781	2.141
DA	1.926	0.846	0.596	1.918	3.294
DD	2.322	0.845	0.995	2.313	3.683

[SAB] Replace with final table of fully paramet-rised SCM

Causes of variation in vaccine responsiveness in a wild
rodent population

Wild vaccinology: determining the causes of variation in
vaccines in a lab-to-wild rodent population

Causes of vaccine failure: bridging the translational
divide using a lab-to-wild model

Causes of low vaccine responsiveness: bridging the
translational divide using a lab-to-wild model

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SUPPLEMENTARY INFORMATION

Supp methods

Insert additional details about the lab mouse colony details of the diet make up of trans-breed and RM1

Supplementary Data: Multilevel Models

Supplementary Data: SCM construction

Supplementary Data: SCM specification

Supplementary Data: Wood mouse data

Supplementary Figures

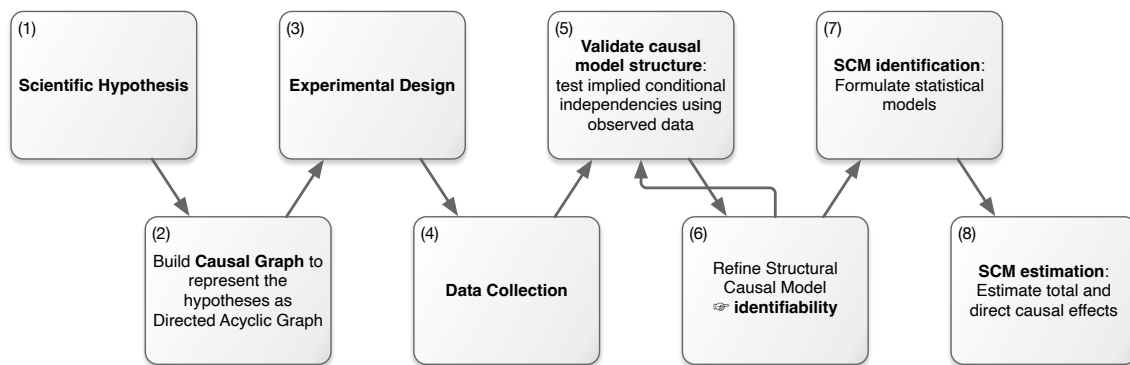


Fig. S1: Flow diagram of the SCM. The Structural Causal Model (SCM) process explicitly represents scientific hypotheses (1) as a directed acyclic graph (DAG). This DAG is mathematically encoded as a set of nested equations that describe the flow of causation between variables, and helps inform lab and field experimental design (3) and data collection (4). After processing, the data are used to test the validity of the causal assumptions underlying the SCM, e.g. by testing the conditional independencies implied by the DAG (5). If the assumptions are not supported, refinements of the SCM (6) are necessary. When all conditions are met, the SCM is identifiable and statistical models can be formulated to adjust for confounding (7). Parameters from these models can then be used as estimates of direct and indirect causal effects (8).

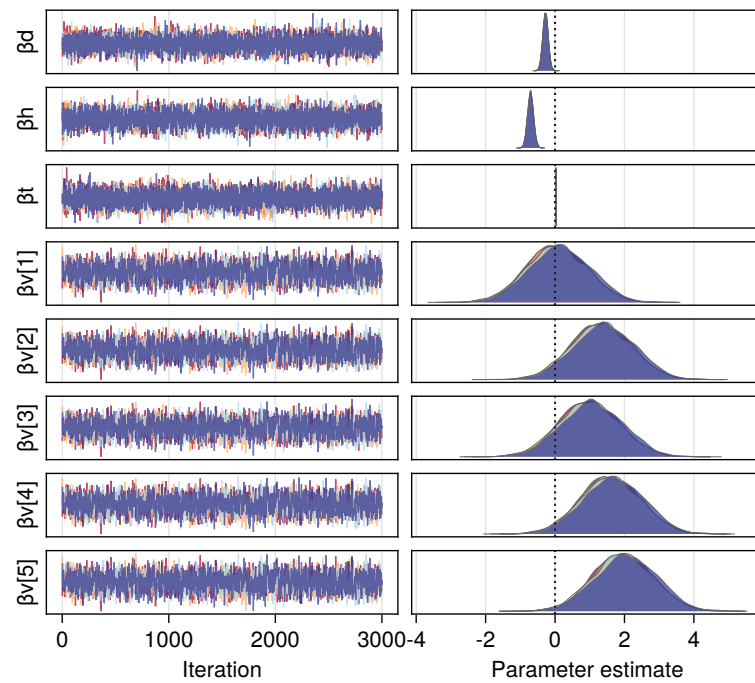


Fig. S2: MCMC chains and posterior distributions of coefficients for Diet, Habitat, Time post immunisation, and vaccine formulations A, AD, D, DA, and DD.